

Supplementary Information for *Conditional validity of quantum event classifiers under collider systematics and quantum estimation uncertainty*

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This supplement gives the full algebra behind the formal results, the registered-falsifier and correction trail, and extended numerical tables. Every numerical table is a transcription or an explicitly identified summary of the immutable JSON artifact named in its caption. Superseded first-run tables remain in the repository under `*_v1_*.json`.

1 Formal details

1.1 Exact weighted anytime-valid certification

Condition on a frozen finite audit population and a scalar $w_{\max}$ fixed before the random audit order. For IID with-replacement labeled draws (c_i, u_i) , with $c_i \in \{0, 1\}$, $u_i \in [0, w_{\max}]$, $E[u] > 0$, and $R = E[uc]/E[u]$, fix $\tau \in [0, 1]$ before observing the audit label stream and define

$$Z_i(\tau) = \frac{u_i(c_i - \tau) + \tau w_{\max}}{w_{\max}}.$$

Because $u_i(c_i - \tau)$ lies between $-\tau w_{\max}$ and $(1 - \tau)w_{\max}$, $Z_i(\tau) \in [0, 1]$. Moreover,

$$E[Z(\tau)] \geq \tau \iff E[u(c - \tau)] \geq 0 \iff E[uc] \geq \tau E[u] \iff R \geq \tau.$$

The reduction is therefore exact. If (L_n, U_n) is any level $1 - \alpha$ confidence sequence for the mean of Z , then on its time-uniform coverage event $L_n \geq \tau$ implies $R \geq \tau$ at every n . Thus

$$P\{\exists n : \text{SUPPORTED and } R < \tau\} \leq \alpha$$

under arbitrary stopping; symmetrically, $P\{\exists n : \text{REFUTED and } R \geq \tau\} \leq \alpha$. Both kinds of incorrect decisive verdict are contained in the same two-sided CS coverage failure, so their union for this fixed claim is also bounded by α . Setting $u \equiv 1$ and $w_{\max} = 1$ recovers the unweighted construction. Weighting changes the variance and hence the stopping time, but adds no validity slack. Labeling reveals (y_i, w_i) and therefore $u_i = w_i$, $w_i 1[y_i = 1]$, or $w_i 1[y_i = 0]$ as appropriate; event-wise weights remain unavailable to I1. The data-curation layer supplies only the single global scalar bound before sampling, not row weights or class labels; this does not enlarge I1 with event-level information. Both classes have positive archived weight mass, establishing $E[u] > 0$ for the class-conditional cases. The multiplier 2.05 covers the largest admissible compound scale $2.0 \times 1.01 = 2.02$.

This guarantee is per fixed claim and simultaneous only over its labeling times/stopping rules. It is not FWER or simultaneous control over thresholds, models, or environments; selecting τ or a claim after labels requires a separate multiplicity construction. Time-uniformity also does not validate arbitrary adaptive row selection. E07 instead uses a score-based proposal fixed before labels and bounded importance weights.

1.2 Information-set boundary for weight-only nuisances

For a weight-only nuisance $P_\theta(X) = P_0(X)$ and an unchanged correctness process. Hence every I1 statistic has the same sampling law under θ and 0, and so does I2 when labels carry nominal weights. Any test of size at most α based on those observations has power equal to its actual size and hence at most α against the nuisance (exactly α only for exact size α). No nontrivial distinction is possible; under our fail-closed policy, absence of distinguishing evidence is reported UNRESOLVED for claims whose truth changes only through hidden rate weights.

At I3 a control-region count $N \sim \text{Poisson}(\lambda(\theta))$ with $\lambda(\theta) \neq \lambda(0)$ changes the sampling law and restores non-trivial power. The restoration is quantitative rather than absolute: template variance

$$\sigma_c^2 = \sum_g (\text{relerr}_g \lambda_g)^2$$

sets the resolution scale. This is why the template-statistics-aware E14 fit identifies the ttbar rate only to approximately $\pm 10\%$ and leaves tighter claims unresolved.

1.3 Random deployments and verdict stability

Let ω denote the finite-shot or hardware realization and $\tilde{f}_\omega$ the resulting refit, recalibration, and threshold. If, conditional on ω , the audit stream remains IID and disjoint from training, calibration, and threshold selection, and the audit controls false certification,

$$P(\text{false certification} \mid \omega) \leq \alpha,$$

then the tower property gives marginal control for both the deployment-relative and ideal-anchored claim classes:

$$P(\text{false certification}) = E_\omega[P(\text{false certification} \mid \omega)] \leq \alpha.$$

For signed target and source movements ΔM_T and ΔM_S , the exact margin movements are ΔM_T for the ideal-anchored claim and $\Delta M_T - \Delta M_S$ for the deployment-relative claim. Therefore $|m^*| > |\Delta M_T|$ or, respectively, $|m^*| > |\Delta M_T - \Delta M_S|$ is sufficient to preserve the truth-sign; failure of a sufficient condition gives no conclusion. On coverage, any decisive verdict points to that sign. Equality of two realized ternary verdicts additionally requires that both audits resolve and that both CSs cover: without joint calibration the intersection probability is at least $1 - 2\alpha$, or $1 - \alpha$ if each audit uses $\alpha/2$. For weighted claims the metric-scale margin is mapped to the certification scale by $E[u]/w_{\max}$. Common-mode movement cancels from the deployment-relative margin but not from the ideal-anchored margin. E16 archives only ΔM_S , not ΔM_T or $\Delta M_T - \Delta M_S$; Figure S16 and Table S6 therefore do not instantiate either condition. The proposition remains conditional, while archived flip rates are independent empirical evidence.

1.4 Pre-split component allocation for weighted balanced accuracy

For $\text{BA}_w = (\text{TPR}_w + \text{TNR}_w)/2 \geq \tau_{\text{BA}}$, the mathematical construction can predeclare component thresholds τ_1, τ_2 satisfying $(\tau_1 + \tau_2)/2 = \tau_{\text{BA}}$ and error levels $\alpha_1 + \alpha_2 = \alpha$. Each component uses the exact one-sample reduction with a valid class-specific bound. The BA claim is SUPPORTED only if both components are supported, REFUTED only if both are refuted, and UNRESOLVED otherwise. If BA is

false, at least one component claim is false; certifying it requires a coverage violation. A union bound therefore controls false certification at α . The symmetric argument controls false refutation. In the executed E13v2 battery, however, the class maxima were calculated using labels from the complete frozen population. It is therefore an oracle/benchmark diagnostic rather than an operational I2 guarantee. Table S9 shows that even this favorable oracle construction is information-limited by the benchmark’s weighted signal fraction; successful TNR resolutions are reported only in that diagnostic sense.

2 Registered falsifiers and audit trail

The count below is by *unique registered falsifier arm*. The E15 calibration gate is one arm, although it rejected two successive invalid implementations before the valid grid was allowed to run. This convention reconciles the nine-arm tally with the two gate interventions.

Table S1: All nine registered falsifier arms that fired. Frozen conditions and consequences are recorded in `docs/experiment_registry.md`.

Arm	Frozen condition	Observed firing	Consequence obeyed
E02R	TES sign pattern fails multi-seed replication	Up-shift monotonicity in only 2/5 seeds	Down-shift retained provisionally within-world; E17 later withdrew the directional cross-world claim.
E12(e)	Two flagship decoupling cells fail to reproduce	Diboson flagship cells stayed nominal-like in the fresh world	Normalization-collapse statement scoped to signal-region composition; failing registered cells kept.
E14 v1	Control-region interval coverage below nominal	Pure-Poisson fit gave ttbar bias +0.07 and zero coverage	Run blocked; template-statistics likelihood registered, then rerun as v2.
E15 gate	Nominal L2 coverage outside 0.6827 ± 0.02	Conditional-ensemble and numerical-conditioning implementations each failed	Both grids blocked; ensemble semantics and optimizer fixed before any shifted claim.
E17(b)	Degradation signs fail across additional worlds	One world improved QK under combo by 0.009; another under TES-down by 0.005	Cross-world degradation claim withdrawn.
E08v2(a)	Corrected counting coverage outside the nominal band	One-draw corrected coverage 0.9916–1.000	Arm blocked and bounded multi-draw E08v3 registered.
E08v2(b)	Independent-template flagship L2 coverage < 0.633	Coverage 0.000 versus shared 0.7188	“Profiling restores validity” made explicitly shared-simulation-conditional.
E08v3(a)	Marginal symmetric-BB coverage outside 0.6827 ± 0.06 for at least two gated models	All three gated models overcovered, 0.7513–0.8284	Calibration-restoration claim withdrawn; conservative direction reported.
E13v2(b)	All true benchmark BA claims unresolved by 5,000 labels	TPR component 0/200 resolutions in every cell	BA path published as information-limited; TNR success and mechanism reported separately.

Table S2: Audit-trail summary. The full findings and dispositions are in `docs/audits/`.

Audit	Scope	Material effect
Pre-campaign, 2026-08-11	All E00–E11 code, tables, splits, and governance	Seven design constraints frozen before E12–E16: archived row indices, role separation, post-selection sensor validation, an independent floor, independent quantum-noise streams, a typed deployment snapshot, and weighted estimands. No prior headline invalidated.
Post-campaign, 2026-08-11	E12–E16, E04v3, E11v2; 172 registry values and about 178 manuscript claims	E13 nominal-weight estimand fixed and rerun; E15 gate rejected two implementations; E16 dual accounting added; E12 normalization reading corrected; all superseded tables retained.
Pre-submission, 2026-08-12	Extension runners, figure scripts, and line-by-line formal proofs	Two HIGH findings fixed: JES tracking scoped and disclosed; E19 weighted estimand corrected and rerun. Proposition 4 gained its coverage-event and scale-bridge qualifiers. Eight LOW findings were fixed, completed, disclosed, or documented.
F8.2, 2026-08-12	Final LaTeX, supplement, number trace, semantic equivalence, rendered PDFs	E08v3 bias range corrected from +6.1 to the archived +7.5 endpoint; the nine-arm counting convention made explicit.
Final mathematical/editorial, 2026-08-12	Proposition 2, Theorem 1, Proposition 4, C2/C3, E17, figures, and three independent adversarial reviews	Claims were restricted to their exact mathematical and evidential scope: per-claim rather than family-wise control; finite-population-conditional weight bounds; conditional Proposition 4 with empirical flips separated; joint nuisance-representability/template-quality limits; no new experiment or result.

3 Extended numerical results

3.1 Inference sensitivity (E15)

Table S3 reports the local slope at the nominal point, averaged over the five injected μ values. L1 is the counting estimator, L2 the complete profile, and L3 the profile with the shifted family omitted. Only the three models that passed the nominal L2 gate are shown. Units are μ per nuisance unit: per standard deviation for TES and JES, per GeV for soft-MET, and per unit scale for the normalization families.

The archived initial global optimizer-success flag is conservative and falls as low as 0.382. Analytic gradients, multi-start fitting, a monotone lower-minimum safeguard, and the nominal coverage gate support the reported solutions but do not prove global optimization in every shifted cell.

Table S3: E15 local sensitivity and nuisance tracking. Source: `results/tables/E15_sensitivity.json`.

Family	Model	L1 slope	L2 slope	L3 slope	L2 tracking
TES	QK-SVC	9.8095	0.4092	7.1213	0.9970
TES	RBF-SVC	10.3065	0.2269	8.6641	0.9986
TES	XGBoost	5.4959	0.1031	4.7948	0.9938
JES	QK-SVC	3.6530	-0.2033	5.1238	0.9813
JES	RBF-SVC	2.7528	-0.1096	5.7069	0.9901
JES	XGBoost	1.7328	0.0121	1.1957	0.7116
soft-MET	QK-SVC	-0.0812	-0.2126	6.5142	0.2497
soft-MET	RBF-SVC	0.7731	-0.0496	3.0473	0.2511

Family	Model	L1 slope	L2 slope	L3 slope	L2 tracking
soft-MET	XGBoost	0.7305	-0.4605	-1.5627	0.5001
ttbar scale	QK-SVC	49.7510	0.5175	-0.7125	1.0040
ttbar scale	RBF-SVC	54.0570	-1.1795	-4.0830	0.9915
ttbar scale	XGBoost	12.7935	-0.0220	9.0305	0.9895
diboson scale	QK-SVC	4.7626	0.0834	0.5676	0.9900
diboson scale	RBF-SVC	4.3356	0.2769	1.0303	0.9869
diboson scale	XGBoost	0.9981	-0.0065	0.4824	1.0006
background scale	QK-SVC	270.1200	-30.9900	222.4700	1.0000
background scale	RBF-SVC	397.2600	34.5600	823.4100	0.9800
background scale	XGBoost	101.8800	-5.3900	-3.6800	1.0000

3.2 Fresh-world validity and stopping-time efficiency

Table S4: Fresh-world validity replication. Source: `results/tables/E19_fresh_world_validity.json`; the false-refutation counts documented in the registered status agree with the rates stored in the table.

Accounting	False certification	False refutation	$\alpha + 3\sigma$ boundary
Unweighted, non-vetoed	11/3,060 = 0.359%	1/11,980 = 0.008%	6.182%
Unweighted, all streams	11/7,700 = 0.143%	1/11,980 = 0.008%	6.182%
Weighted, nominal weights	6/7,980 = 0.075%	4/11,700 = 0.034%	5.732%

All 12 archived model–environment score comparisons were byte-identical before auditing. The weighted bound was $w_{\max} = 7.22421 \times 2.05 = 14.80963$. The corrected nominal-weight estimand is invariant across all 12 weight-only environments. These correlated Monte-Carlo rates are implementation stress tests, not a proof of the theorem or family-wise error control across the claim grid.

Table S5: Stopping-time price relative to the Wald information yardstick. Source: `results/tables/E06_nstar_efficiency.json`.

Subset	Cells	Q25	Median	Q75
All resolved cells	518	1.56	2.07	2.97
Margin $[0.01, 0.02)$	95	2.82	3.35	3.85
Margin $[0.02, 0.04)$	95	2.62	3.19	3.66
Margin $[0.04, 0.08)$	164	1.63	1.98	2.38
Margin ≥ 0.08	164	1.25	1.46	1.72
Certification direction	487	1.54	1.99	2.78
Refutation direction	31	3.12	3.34	3.65

3.3 Quantum-estimation diagnostics (E16)

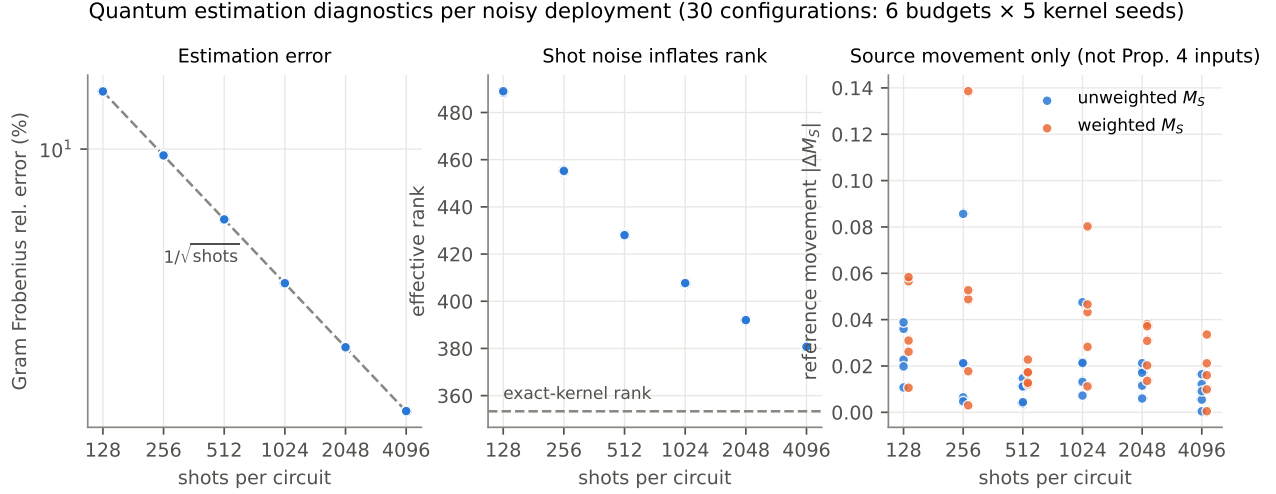


Figure S16: Per-configuration diagnostics for all 30 noisy deployments: relative Frobenius error, effective-rank inflation, and source-reference movement $|\Delta M_S|$. The target movement ΔM_T and the difference $\Delta M_T - \Delta M_S$ were not archived, so this panel cannot instantiate Proposition 4. Source: `results/tables/E16_quantum_uncertainty.json`.

Table S6: E16 shot-budget summary. Frobenius error and effective rank are means over five deployments; movement is only the maximum absolute weighted source-reference shift $|\Delta M_S|$, not either quantity needed by Proposition 4. Flip and abstention entries are independent empirical percentages. Source: `results/tables/E16_quantum_uncertainty.json`.

Shots	Frob. err. (%)	Rank	max $ \Delta M_S $	Far flips own	Far flips fixed	Moderate flips own	Moderate flips fixed	Near abst. own
128	13.66	488.6	0.058	0.0	20.8	15.8	71.4	94.0
256	9.66	455.2	0.139	0.0	17.7	16.1	49.4	92.9
512	6.84	428.3	0.023	0.0	0.9	11.5	47.7	92.5
1024	4.83	407.6	0.080	0.0	11.9	13.2	54.6	92.0
2048	3.42	392.1	0.038	0.0	5.8	11.1	58.4	93.9
4096	2.42	380.7	0.034	0.0	0.4	6.7	40.4	93.0

3.4 Independent-MC split studies (E08v3)

Table S7: Marginal nominal counting coverage over 400 independent splits. Source: `results/tables/E08v3_multidraw.json`.

Model	Shared truth	Indep. naive	Indep. BB	Indep. BB symmetric	Stored prediction
QK-SVC	0.6828	0.0691	0.6501	0.8113	0.5220
RBF-SVC	0.6827	0.0860	0.6605	0.8284	0.5220
XGBoost (A)	0.6830	0.0673	0.5650	0.7513	0.5216
XGBoost (B)	0.6823	0.0645	0.5458	0.7033	0.5164

Table S8: Independent-template fixed-profile results in the two registered E08v3 cells across 10 splits, averaged over the frozen μ grid within each split. This does not test MC-statistics-aware profiling in general. Source: `results/tables/E08v3_multidraw.json`.

Cell	Coverage range	Draws < 0.633	Bias range	Shared reference
TES = 0.98, XGBoost	0.000–0.238	10/10	−6.600–+4.050	0.7188
Nominal, XGBoost	0.000–0.359	10/10	−6.557–+7.496	gate-passing in E15

3.5 Weighted balanced-accuracy allocation (E13v2)

Table S9: E13v2 oracle component-allocation battery, 200 replications per cell at $n_{\max} = 5,000$. Class bounds use full-population labels, so this is not an operational I2 guarantee. Entries are correct resolutions; all false-certification counts are zero. Source: `results/tables/E13v2_baw_allocation.json`.

Population	Claim	BA pre-split	TPR component	TNR component	v1 BA	median n^*
Synthetic	$m = 0.02$, true	0/200	9/200	36/200	0/200	–
Synthetic	$m = 0.02$, false	0/200	3/200	36/200	0/200	–
Synthetic	$m = 0.05$, true	83/200	83/200	199/200	0/200	3,489
Synthetic	$m = 0.05$, false	87/200	87/200	200/200	0/200	3,370
Benchmark	$m = 0.02$, true	0/200	0/200	67/200	0/200	–
Benchmark	$m = 0.02$, false	0/200	0/200	64/200	0/200	–
Benchmark	$m = 0.05$, true	0/200	0/200	200/200	0/200	–
Benchmark	$m = 0.05$, false	0/200	0/200	200/200	0/200	–

Table S10: Mechanism diagnostics for E13v2 at $n = 5,000$.

Diagnostic	Synthetic control	Benchmark physics weights
Weighted signal fraction	0.301459	0.000966
Positive-class ESS ratio	0.226298	0.001106
$w_{\max}^{(+)}$	4.58547	5.98645
$w_{\max}^{(-)}$	9.40021	14.89144
TPR Z -margin at $m = 0.05$	0.01130253	0.00002827
TPR CS radius	0.011776	0.001806
v1 BA radius	0.168819	0.288855

In the benchmark, the TPR radius-to-margin ratio is about 64 at 5,000 labels. A quadratic information-scaling extrapolation therefore gives $n^* \approx 5,000 \times 64^2 \simeq 2 \times 10^7$ uniform labels at margin 0.05. This is a diagnostic extrapolation, not a new Monte Carlo run.

4 Artifact map

The main machine-readable sources for this supplement are:

- `results/tables/E15_sensitivity.json`;
- `results/tables/E19_fresh_world_validity.json`;
- `results/tables/E06_nstar.json` and `results/tables/E06_nstar_efficiency.json`;
- `results/tables/E16_quantum_uncertainty.json`;
- `results/tables/E08v2_independent_mc.json` and `results/tables/E08v3_multidraw.json`;
- `results/tables/E13v2_baw_allocation.json`.

The frozen protocols and post-run dispositions are in `docs/experiment_registry.md`, `docs/decisions.md`, and the three dated files under `docs/audits/`. These text records explain the design and corrections; the JSON artifacts remain the numerical source of truth.